Viren Kumar

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OBJECTIVE

Undergraduate in Computer Engineering and Applied Mathematics seeking higher level work with Markov Chains, Mathematical Modeling, and Systems Optimization in applied problems.

EDUCATION

California State Polytechnic University, Pomona **August 2024 – May 2027 expected**

Dual Major: Bachelor of Science, Computer Engineering
Bachelor of Science, Applied Mathematics

RESEARCH AND TECHNICAL EXPERIENCE

McNair Scholar and Undergraduate Researcher **August 2025 -- Present**

McNair Scholars Program | Cal Poly Pomona Pomona, CA

- Conducted research under Dr. Aly on post-quantum encryption for CubeSat communication as part of the NASA ORBIT competition.
- Focused on structural and mathematical validation of Applying ASCON to the Kyber protocols, including using group theory to verify monoid behavior and lattice-like structure.
- Phase III contestant with team presenting work at Johnson Space Center in Houston.

CubeSTEP Flight Software Lead **December 2025 -- Present**

CubeSTEP | Cal Poly Pomona Pomona, CA

- Collaborate with the JPL F' team to develop flight-ready software for an incoming satellite mission.
- Co-author of paper and presented poster on "Mission-Oriented Flight Software Development: CubeSTEP," for SMC-IT 2026.
- Promoted to software lead due to exceptional performance on August 2026

Research Volunteer **June 2025 -- August 2025**

NCUHS | UCLA Los Angeles, CA

- Collaborated with Dr. Guido Faas on fabrication of micro-sensors and chips for implantation in mice for pain and migraine research.
- Supported chip handling, microfabrication preparation, and technical lab workflows in a biomedical research setting.

Founding Member and Thermal Engineering Lead **August 2024 -- December 2025**

CPP Smartphone Project | Cal Poly Pomona Pomona, CA

- Helped establish project scope and secure funding for a student-led effort to design a smartphone system using FPGA prototyping.
- Led a five-person thermal engineering team focused on body design, thermal management, and sustained performance.
- Coordinated with hardware, system-on-chip, and systems teams to manage thermal and energy budgets; used FreeCAD and Ansys for design iteration.

CPP Problem Solving Team Member **Spring 2024 -- Present**

Problem Solving Team | Cal Poly Pomona Pomona, CA

- Solve problems from mathematical journals and prepared formal solutions in LaTeX for submission.
- Personally submitted verified solutions to CRUX, PME, and CMJ.
- Became main point of contact for proof-writing and inequality problem solving.

Materials Engineering Intern **June 2022 -- August 2022**

LyTen Inc. San Jose, CA

- worked on creating and testing various composites impregnated with different variations of Graphene nanoplateles
- Through labwork formalized and presented a method to filter for grain size with a 70% higher yeild, data was organized and calculated through matlab and excel and presented via powerpoint.

PUBLICATIONS AND MANUSCRIPTS

Mission-Oriented Flight Software Development: CubeSTEP **August 5, 2026**

- 2nd Co-author; publication for SMC-IT 2026.

From TinyML Model to Open Silicon, A CFU Playground Pipeline for AI design on GF180MCU **August 12, 2026**

- 1st Author; publication for STARS 2026.

Ascon-Kyber for Quantum-Resilient Space Telemetry **May 22, 2026**

- 1st author; publication for McNair Symposium 2026.

PRESENTATIONS, PANELS, AND SELECTED TALKS

Generalizations on Markov Chain Steady State Behavior **Nov 5 2026**

- Presented research and progress on various steady state observations Markov Chains, Presented stepsize observation at MAA Southern California-Nevada Section Fall Meeting then proof for related conjectures at 2026 AMS Fall Wester Sectional Meeting, and future progress at the JMM 2027.

MDO to Software Deign **August 5 2026**

- Presented poster at SMC-IT with same name as submitted paper. Focused on methodology for turning MDO(Mission Design Optimizations) into actionable software procedures.

Markov Processes and Weak Stationarity **April 15, 2026**

- Presented a lemma showing that a Markov process at steady state is weakly stationary. At Cal Poly Pomona's 2026 Science Symposium.

Quaterions: An introduction of Abstract Structures **April 8, 2026**

- Presented a poster on Cayley–Dickson constructions at SIAM 2026 Symposium at Cal Poly Pomona

AI Subject Matter Expert Panelist **June 4, 2025**

- Panelist as an AI subject matter expert for the OAI Faculty and Staff Summer Conference for the Cal Poly Pomona Student Perspectives on the AI-Driven Future of Education and Work for STEM to Social Science Fields event.

Thermal Design Optimization for Mobile Devices **August 7, 2025**

- Presented research proposal at CARS Conference, Cal Poly Pomona.

Negative Probabilites in Markov Chains **October 27, 2024**

- Presented Dr. Krinik's research at the Fall Western Sectional Meeting, University of California, Riverside.

OTHER CONFERENCES AND WORKSHOPS

- Forum for Diversity in Graduate Education, University of California, SCSU, October 3, 2026
- 12th International Conference on Space Mission Challenges for Information Technology (SMC-IT),August 3–6 2026
- 5th Annual Quantum Engineering Workshop, University of Southern California, May 29, 2025
- IPAM PUMA: Practicum for Undergraduate Mathematicians in Topology, UCLA, April 12–13, 2025
- Forum for Diversity in Graduate Education, University of California, Riverside, October 19, 2024
- 4th Annual Quantum Engineering Workshop, University of Southern California, May 30, 2024

SELECTED COURSEWORK

Mathematics and Statistics

- MAT 5300: Random Processes
- MAT 4010: Numerical Analysis I
- MAT 4190: Advanced Linear Algebra
- MAT 4170: Abstract Algebra I
- MAT 3470: Combinatorics
- MAT 3140: Real Analysis I
- MAT 4990: Introduction to Mathematical Research with negative states in Markov chains

Electrical and Computer Engineering

- ECE 4300: Computer Architecture
- ECE 3300/L: Digital Circuit Design with Verilog
- ECE 2200/L: Microelectronic Circuits
- ECE 3101/L: Signals and Systems
- ECE 2300/L: Digital Logic Design
- ECE 2101/L: Electrical Circuit Analysis II
- ECE 4990: Engineering Research: Computer Vision
- ECE 4990: Engineering Research: Wireless and Mobile Systems

MEMBERSHIPS, AFFILIATIONS, AND SERVICE

- McNair Scholars Program
- Engineering BRIDGE
- SIAM Vice President and Communications Liasion
- RAMP: Reading, Advising, and Mentoring Program
- MEP-WISE: Maximizing Engineering Potential
- CPP branch IEEE Secretary